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[22EC513]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

CONCEPTS OF ARTIFICIAL INTELLIGENCE AND ITS APPLICATIONS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define PEAS with examples.	L1	CO1	**02**
	b.	List the varieties of constrains with examples.	L1	CO2	**02**
	c.	Define planning in artificial intelligence.	L1	CO3	**02**
	d.	What are the applications of Machine Learning?	L1	CO4	**02**
	e.	Define Neural Network.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the history of an artificial intelligence.	L2	CO1	**05**
		OR			
	b.	Describe the depth first search algorithm with neat sketch.	L2	CO1	**05**
3.	**a.**	Explain the hill climbing algorithm.	L2	CO2	**05**
		OR			
	b.	Describe about the simulated annealing with state space diagram.	L2	CO2	**05**
4.	**a.**	Explain the partial order planning in artificial intelligence.	L2	CO3	**05**
		OR			
	b.	Describe the planning used CSP in artificial intelligence.	L2	CO3	**05**
5.	**a.**	Explain how the Linear regression work in machine learning.	L2	CO4	**05**
		OR			
	b.	Explain the logistic regression model helps in the prediction of machine learning.	L2	CO4	**05**
6.	**a.**	Explain different types of back propagation networks	L2	CO5	**05**
		OR			
	b.	Explain the architecture of pre trained CNN Models	L2	CO5	**05**
7.	**a.**	Explain the uniform cost algorithm with neat diagram.	L2	CO1	**05**
		OR			
	b.	Explain properties of environment in Artificial Intelligence.	L2	CO1	**05**

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		PART-C	**6 X 10=60**		
8.	**a.**	Describe in detail about the various agent types in artificial intelligence.	L2	CO1	**10**
		OR			
	b.	Describe the greedy search algorithm with an example.	L2	CO1	**10**
9.	**a.**	Explain the map coloring of constraint stratification problem with neat sketch.	L2	CO2	**10**
		OR			
	b.	Describe the genetic algorithm with neat diagram.	L2	CO2	**10**
10.	**a.**	Explain the forward search space planning algorithm with neat diagram.	L2	CO3	**10**
		OR			
	b.	Explain the Backward search space planning algorithm with neat diagram.	L2	CO3	**10**
11.	**a.**	Distinguish between supervised and unsupervised machine learning models.	L3	CO4	**10**
		OR			
	b.	Explain in detail about the decision tree with examples.	L2	CO4	**10**
12.	**a.**	Explain in detail about feedforward neural networks with neat sketch.	L2	CO5	**10**
		OR			
	b.	Explain the working principle of a convolutional neural network, and discuss the different building blocks used to extract features from input data.	L2	CO6	**10**
13.	**a.**	Describe the domain splitting and variable elimination.	L2	CO2	**10**
		OR			
	b.	Explain A* search algorithm with examples	L2	CO1	**10**